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PHY101 Quiz 2 Solved

Prepared by VU Answer

1) **Power is equal to the dot product of force and:**

Select the correct option

a) Position vector

b) Displacement

c) Velocity

d) Acceleration

2) **The frequency which is not audible to the human ear is**

Select the correct option

a) 50000 Hz

b) 500 Hz

c) 5000 Hz

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d) 50 Hz

3) **A wheel of radius 50 cm having the angular speed of 5 rad/s will have linear speed in m/s?**

Select the correct option

a) 1.5

b) 2.5

c) 3.5

d) 0.5

4) **When breaks are applied to a fast-moving car, the passenger will be throws:**

Select the correct option

a) Backward

b) Forward

c) Upward

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d) Downward

5) **Stress can be measured in:**

Select the correct option

a) N.m

b) N/m2

c) N.m2

d) N/m

6) **A heavy particle moving with 5 m/s suffers an elastic collision with a light particle at rest. After collision, velocity of light particle will be:**

Select the correct option

a) 10 m/s

b) 2.5 m/s

c) 20 m/s

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d) 5 m/s

6) **Acceleration in a body is always produced on the direction of:**

Select the correct option

a) Velocity r

b) Weight

c) Acceleration

d) Force

7) **The fundamental dimensions of angular momentum are**

Select the correct option

a) ML-2T-2

b) ML2T-1

c) ML2T-2

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d) MLT-1

8) **Ali wants to lift a mass of 7.5 kg with constant velocity by a rope that passes through a frictionless pulley which is attached to the celling of room. Calculate the tension in the rope by neglecting the mass of the rope.**

Select the correct option

a) 75 N

b) 7.5 Kg

c) 75 Kg

d) 0.75 N

9) **The first condition of equilibrium implies that:**

Select the correct option

a) ∑Fy = 0

b) ∑Fx = ∑Fy

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c) ∑F = 0

d) ∑Fx = 0

10) **Final take off velocity of an airplane is 67 m/s. The length of runway is 2 km, the constant acceleration is:**

Select the correct option

a) 3.24 ms^-1

b) 3.24 ms^-2

c) 2.24 ms^-1

d) 2.24 ms^-2

11) **The ultimate strength of a sample in the stress at which the simple:**

Select the correct option

a) remain underwater

b) breaks

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c) bends 180?

d) returns to its original shape when the stress is moved

12) **The area under the velocity time graphs is:**

Select the correct option

a) Acceleration

b) Torque

c) Distance

d) Force

13) **if you travelled for 2.5 hours with an average speed 48 miles/hours, the distance travelled is:**

Select the correct option

a) 19.2 miles

b) 120 miles

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c) 300 miles

d) 48 miles

14) **The law of inertia was firstly formulated by:**

Select the correct option

a) Newton

b) Einstein

c) Galileo

d) Aristotle

15) **When the velocity of an aeroplance is doubled, the momentum ___.**

Select the correct option

a) Is conserved

b) Becomes zero

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c) Increase uniformly

d) Remains unchanged

16) **As we move above, the body above the surface of the earth, the change in potential energy will be:**

Select the correct option

a) Infinity

b) Positive

c) Zero

d) Negative

17) **The slope of a velocity – time the graph at any point may be identified with:**

Select the correct option

a) instantaneous acceleration

b) average acceleration

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c) instantaneous velocity

d) instantaneous acceleration

18) **A body of mass 2 kg moving with velocity 4 m/s has K.E equals to:**

Select the correct option

a) 16 J

b) 8 J

c) 32 J

d) 2 J

19) **A ___ velocity is obtained by dividing the vector by its magnitude:**

Select the correct option

a) negative

b) unit

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c) position

d) normal

20) **A particle oscillating in simple harmonic motion is:**

Select the correct option

a) in equilibrium at the center of its path because the acceleration is zero there

b) never in equilibrium because it is in motion

c) in equilibrium at the end of its path because the acceleration is zero there

d) never in equilibrium because there is always a force

21) **Acceleration of an object must be zero at a point where:**

Select the correct option

a) the instantaneous velocity is not zero but changing

b) the instantaneous velocity is constant

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c) the average velocity is zero

d) the instantaneous velocity is zero but changing

22) **Young's modulus is a proportionality constant that relates the force per unit area applied perpendicularly at the surface of an object to:**

Select the correct option

a) the fractional change in volume

b) the fractional change in length

c) the shear

d) the pressure

23) **A bullet shoot straight up returns to its sharing point in 10 sec. the initial speed was:**

Select the correct option

a) 49 m/s

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b) 9.8 m/s

c) 98 m/s

d) 24.5 m/s

24) **An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:**

Select the correct option

a) 1.57 rad/s

b) 12.6 rad/s

c) 2.0 rad/s

d) 6.3 rad/s

25) **The dimensional units of ratio of work and power is:**

Select the correct option

a) F

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b) J

c) T

d) L

26) **The newton-second is a unit of:**

Select the correct option

a) power

b) angular momentum

c) linear momentum

d) work

27) **A sound wave has a wavelength of 3.0m. The distance from a compression center to the adjacent rarefaction center is:**

Select the correct option

a) 3.0m

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b) 1.5m

c) 0.75m

d) need to know wave speed

28) **For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:**

Select the correct option

a) the center of gravity

b) the same point

c) the geometrical center

d) the center of mass

29) **If a simple pendulum oscillates with an amplitude 50 mm and time period 2s, then its maximum velocity is**

Select the correct option

a) 0.8 m/s

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b) 0.16 m/s

c) 0.15 m/s

d) 0.1 m/s

30) **bank and forth is a leftward and rightward direction. This type of wave is known as a ___.**

Select the correct option

a) Mechanical

b) Electromagnetic

c) Transverse

d) Longitudinal

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